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Candidate surname	Other names
USMAN	Grade A* Scholar
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Mentora Academy Internal Candidate Ref: MEN-EDX-2026-U4-W07	

Pearson Edexcel International Advanced Level

Chemistry

International Advanced Level

UNIT 4: Rates, Equilibria and Further Organic Chemistry

WEEK 7 ASSESSMENT: TOPIC 13A.1 & 13A.2 CHEMICAL EQUILIBRIA (K_c & K_p)

Assessment Paper:	WCH14/01 (Week 7 In-Depth Mastery)
Time Allowed:	2 hours 30 minutes
Total Marks:	150 Marks
Topics Examined:	Topic 13A.1: Equilibrium Constant K _c (Expressions, Units, ICE Calculations) Topic 13A.2: Equilibrium Constant K _p (Partial Pressures, Mole Fractions, K _p Calculations)

INSTRUCTIONS TO CANDIDATES

- Use black ink or ball-point pen. Fill in the boxes at the top of this page.
- Answer ALL questions in Section A, Section B, and Section C.
- In Section A (Questions 1 to 30), select your answer by placing a clear cross in the box [X].
- In Section B and C, show all working in calculation questions. Include appropriate units where required.
- Questions marked with an asterisk (*) are those where your quality of written communication and logical sequence of structure will be assessed.
- A Periodic Table and physical constants are printed on the back page of this booklet.

INFORMATION FOR CANDIDATES

- The total mark for this paper is 150.
- Section A: 30 marks; Section B: 90 marks; Section C: 30 marks.
- Full worked confidential teacher mark scheme is provided at the rear for post-assessment diagnostic.

SECTION A

Answer ALL questions. For each question, make a selection by putting a cross in one box [X]. If you change your mind, put a line through the box and then mark your new choice with a cross.

1 What are the correct units of the equilibrium constant K_c for the reaction: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$?

- ☐ **A** mol dm^{-3}
☐ **B** $\text{dm}^3 \text{mol}^{-1}$
☐ **C** $\text{dm}^6 \text{mol}^{-2}$
☐ **D** No units (dimensionless)

(Total for Question 1 = 1 mark)

2 Which is the correct expression for K_c for the heterogeneous equilibrium: $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$?

- ☐ **A** $K_c = [\text{CaO}][\text{CO}_2] / [\text{CaCO}_3]$
☐ **B** $K_c = [\text{CO}_2]$
☐ **C** $K_c = 1 / [\text{CO}_2]$
☐ **D** $K_c = [\text{CaO}] / [\text{CaCO}_3]$

(Total for Question 2 = 1 mark)

3 A gas mixture contains 0.20 mol of N_2 , 0.60 mol of H_2 , and 0.20 mol of NH_3 at a total pressure of 5.0 atm. What is the partial pressure of H_2 in the mixture?

- ☐ **A** 0.60 atm
☐ **B** 1.50 atm
☐ **C** 3.00 atm
☐ **D** 0.30 atm

(Total for Question 3 = 1 mark)

4 What are the correct units of K_p for the reaction: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$, when partial pressures are measured in kPa?

- ☐ A kPa
- ☐ B kPa^{-1}
- ☐ C kPa^{-2}
- ☐ D Dimensionless

(Total for Question 4 = 1 mark)

5 For the equilibrium: $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$, why is the value of K_c identical to the value of K_p at any given temperature?

- ☐ A All reactants and products are non-polar diatomic gases
- ☐ B The total number of moles of gas is the same on both sides of the equation ($n_{\text{gas}} = 0$)
- ☐ C The activation energy of the forward reaction equals the activation energy of the reverse reaction
- ☐ D The reaction occurs without any enthalpy change ($H = 0$)

(Total for Question 5 = 1 mark)

6 In the esterification reaction: $\text{CH}_3\text{COOH}(\text{l}) + \text{C}_2\text{H}_5\text{OH}(\text{l}) \rightleftharpoons \text{CH}_3\text{COOC}_2\text{H}_5(\text{l}) + \text{H}_2\text{O}(\text{l})$, why can equilibrium moles be used directly in the K_c expression without dividing by the total volume V ?

- ☐ A All substances are organic liquids with identical densities
- ☐ B The total volume V appears to the same power in the numerator and denominator, canceling out completely
- ☐ C Water acts as a solvent and its concentration is therefore constant
- ☐ D The reaction is heterogeneous and liquids are omitted from K_c

(Total for Question 6 = 1 mark)

7 At 500 K, the equilibrium constant K_c for the dissociation: $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ is $0.040 \text{ mol dm}^{-3}$. If the equilibrium concentrations of PCl_3 and Cl_2 are both 0.10 mol dm^{-3} , what is the equilibrium concentration of PCl_5 ?

- ☐ A $0.0040 \text{ mol dm}^{-3}$
- ☐ B 0.25 mol dm^{-3}
- ☐ C 0.40 mol dm^{-3}
- ☐ D 2.50 mol dm^{-3}

(Total for Question 7 = 1 mark)

8 A sealed 2.0 dm^3 container contains 0.40 mol of NO_2 and 0.80 mol of N_2O_4 at equilibrium: $2\text{NO}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}_4(\text{g})$. What is the value of K_c in $\text{dm}^3 \text{ mol}^{-1}$?

- ☐ A 2.0
- ☐ B 5.0
- ☐ C 10.0
- ☐ D 20.0

(Total for Question 8 = 1 mark)

9 The total pressure of an equilibrium mixture of $\text{N}_2\text{O}_4(\text{g})$ and $\text{NO}_2(\text{g})$ is 1.50 atm . If the partial pressure of N_2O_4 is 0.50 atm , what is the value of K_p for $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$?

- ☐ A 0.50 atm
- ☐ B 1.00 atm
- ☐ C 2.00 atm
- ☐ D 4.00 atm

(Total for Question 9 = 1 mark)

10 Which expression correctly defines the mole fraction, x_A , of component A in a gaseous mixture of A, B, and C?

- ☐ A $x_A = p_A / P_{\text{total}}$
- ☐ B $x_A = n_A / (n_A + n_B + n_C)$
- ☐ C $x_A = [A] / ([A] + [B] + [C])$
- ☐ D Both A and B are correct

(Total for Question 10 = 1 mark)

11 For the Haber process: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$, the expression for K_p in terms of partial pressures is:

- ☐ A $K_p = p(\text{N}_2) * p(\text{H}_2)^3 / p(\text{NH}_3)^2$
- ☐ B $K_p = p(\text{NH}_3)^2 / [p(\text{N}_2) * p(\text{H}_2)^3]$
- ☐ C $K_p = 2p(\text{NH}_3) / [p(\text{N}_2) + 3p(\text{H}_2)]$
- ☐ D $K_p = p(\text{NH}_3) / [p(\text{N}_2) * p(\text{H}_2)]$

(Total for Question 11 = 1 mark)

12 Solid carbon reacts with carbon dioxide gas at 1000 K: $\text{C}(\text{s}) + \text{CO}_2(\text{g}) \rightleftharpoons 2\text{CO}(\text{g})$. What is the expression for K_p ?

- ☐ A $K_p = p(\text{CO})^2 / [p(\text{CO}_2) * p(\text{C})]$
- ☐ B $K_p = p(\text{CO})^2 / p(\text{CO}_2)$
- ☐ C $K_p = p(\text{CO}) / p(\text{CO}_2)$
- ☐ D $K_p = 2p(\text{CO}) / p(\text{CO}_2)$

(Total for Question 12 = 1 mark)

13 At a certain temperature, 1.0 mol of H_2 and 1.0 mol of I_2 are placed in a 1.0 dm^3 flask and allowed to reach equilibrium: $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$. If $K_c = 64$, how many moles of HI are present at equilibrium?

- ☐ A 0.80 mol
- ☐ B 1.60 mol
- ☐ C 0.89 mol
- ☐ D 1.78 mol

(Total for Question 13 = 1 mark)

14 How does adding a solid catalyst affect the numerical value of K_c for a reversible reaction?

- ☐ A Increases K_c by lowering the activation energy of the forward reaction
- ☐ B Decreases K_c because products form more rapidly
- ☐ C Has no effect on the numerical value of K_c
- ☐ D Increases K_c only if the reaction is endothermic

(Total for Question 14 = 1 mark)

15 An equilibrium mixture for: $2\text{A}(\text{g}) \rightleftharpoons \text{B}(\text{g})$ has $p_A = 20 \text{ kPa}$ and $p_B = 80 \text{ kPa}$. What is the value of K_p ?

- ☐ A 0.20 kPa^{-1}
- ☐ B 4.0 kPa^{-1}
- ☐ C 0.20 kPa
- ☐ D 0.005 kPa^{-1}

(Total for Question 15 = 1 mark)

16 For the reaction: $\text{CO(g)} + 2\text{H}_2\text{(g)} \rightleftharpoons \text{CH}_3\text{OH(g)}$, what are the units of K_p if partial pressures are measured in bar?

- ☐ A bar
- ☐ B bar^{-1}
- ☐ C bar^{-2}
- ☐ D Dimensionless

(Total for Question 16 = 1 mark)

17 In an experiment, 2.0 mol of SO_2 and 1.0 mol of O_2 are mixed. At equilibrium, 1.6 mol of SO_3 has formed: $2\text{SO}_2\text{(g)} + \text{O}_2\text{(g)} \rightleftharpoons 2\text{SO}_3\text{(g)}$. How many moles of O_2 remain at equilibrium?

- ☐ A 0.20 mol
- ☐ B 0.40 mol
- ☐ C 0.60 mol
- ☐ D 0.80 mol

(Total for Question 17 = 1 mark)

18 The decomposition of ammonium hydrogen sulfide: $\text{NH}_4\text{HS(s)} \rightleftharpoons \text{NH}_3\text{(g)} + \text{H}_2\text{S(g)}$ reaches equilibrium in an evacuated flask at 25 °C with a total pressure of 0.60 atm. What is the value of K_p ?

- ☐ A 0.090 atm^2
- ☐ B 0.36 atm^2
- ☐ C 0.30 atm^2
- ☐ D 0.60 atm^2

(Total for Question 18 = 1 mark)

19 Which change will cause the value of the equilibrium constant K_p to change?

- ☐ A Increasing the total pressure of the gas mixture
- ☐ B Increasing the temperature of the system
- ☐ C Adding an inert gas at constant volume
- ☐ D Adding a powdered heterogeneous catalyst

(Total for Question 19 = 1 mark)

20 For the reaction: $A(g) + B(g) \rightleftharpoons C(g) + D(g)$, $K_c = 16$ at 300 K. A mixture containing 1.0 mol of each of A, B, C, and D is placed in a 1.0 dm³ flask. In which direction will the reaction proceed?

- ☐ A To the right (forward), because the reaction quotient $Q < K_c$
- ☐ B To the left (reverse), because the reaction quotient $Q > K_c$
- ☐ C The system is already at dynamic equilibrium
- ☐ D The reaction will stop immediately

(Total for Question 20 = 1 mark)

21 What is the relationship between K_p and K_c for a gas phase reaction?

- ☐ A $K_p = K_c \cdot (RT)^{\Delta n}$
- ☐ B $K_p = K_c / (RT)^{\Delta n}$
- ☐ C $K_p = K_c \cdot R \cdot T$
- ☐ D $K_p = K_c / (R \cdot T)$

(Total for Question 21 = 1 mark)

22 When 0.10 mol of PCl_5 is heated in a 1.0 dm^3 vessel, it dissociates according to: $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$. At equilibrium, 0.040 mol of Cl_2 is present. What is the value of K_c in mol dm^{-3} ?

- ☐ A 0.016
- ☐ B 0.027
- ☐ C 0.040
- ☐ D 0.060

(Total for Question 22 = 1 mark)

23 If the equilibrium constant for: $2\text{NO}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}_4(\text{g})$ is K_{c1} , what is the equilibrium constant K_{c2} for the reverse reaction: $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$?

- ☐ A $K_{c2} = -K_{c1}$
- ☐ B $K_{c2} = 1 / K_{c1}$
- ☐ C $K_{c2} = \sqrt{K_{c1}}$
- ☐ D $K_{c2} = (K_{c1})^2$

(Total for Question 23 = 1 mark)

24 A vessel contains an equilibrium mixture of 0.30 mol CO , 0.10 mol H_2O , 0.60 mol CO_2 , and 0.60 mol H_2 : $\text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}_2(\text{g}) + \text{H}_2(\text{g})$. What is the numerical value of K_c ?

- ☐ A 2.0
- ☐ B 6.0
- ☐ C 12.0
- ☐ D 18.0

(Total for Question 24 = 1 mark)

25 In the reaction: $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$, the degree of dissociation is and the initial amount of N_2O_4 is 1 mol. What is the total number of moles of gas at equilibrium?

- ☐ A 1 -
- ☐ B 1 +
- ☐ C 1 + 2
- ☐ D 2

(Total for Question 25 = 1 mark)

26 Under what conditions does the addition of an inert gas (such as argon) shift the position of an equilibrium at constant temperature?

- ☐ A When added at constant volume to a reaction with $n_{\text{gas}} = 0$
- ☐ B When added at constant pressure to a reaction where n_{gas} is not equal to 0
- ☐ C When added at constant volume to any reaction
- ☐ D An inert gas can never affect equilibrium position under any circumstances

(Total for Question 26 = 1 mark)

27 For the reaction: $2\text{A}(\text{g}) + \text{B}(\text{g}) \rightleftharpoons 2\text{C}(\text{g})$, all gases are at partial pressures of 2.0 bar at equilibrium. What is the value of K_p ?

- ☐ A 0.25 bar^{-1}
- ☐ B 0.50 bar^{-1}
- ☐ C 1.00 bar^{-1}
- ☐ D 2.00 bar^{-1}

(Total for Question 27 = 1 mark)

28 Why are pure liquids and solids assigned an activity of 1 (and hence omitted) in equilibrium constant expressions?

- ☐ A Their concentrations are virtually zero compared to gases
- ☐ B Their density is constant at a given temperature, so their concentration / active mass remains constant
- ☐ C They do not collide with reactant molecules
- ☐ D They have zero entropy at equilibrium

(Total for Question 28 = 1 mark)

29 At 1000 K, K_p for the reaction: $\text{C(s)} + \text{H}_2\text{O(g)} \rightleftharpoons \text{CO(g)} + \text{H}_2\text{(g)}$ is 1.60 atm. If the equilibrium partial pressure of steam is 0.40 atm and $p(\text{CO}) = p(\text{H}_2)$, what is the partial pressure of CO?

- ☐ A 0.40 atm
- ☐ B 0.64 atm
- ☐ C 0.80 atm
- ☐ D 1.26 atm

(Total for Question 29 = 1 mark)

30 Which statement correctly describes what happens when the volume of the reaction vessel is doubled for: $2\text{SO}_2\text{(g)} + \text{O}_2\text{(g)} \rightleftharpoons 2\text{SO}_3\text{(g)}$ at constant temperature?

- ☐ A K_c decreases and equilibrium shifts to the left
- ☐ B K_c increases and equilibrium shifts to the right
- ☐ C K_c remains constant and equilibrium shifts to the left
- ☐ D K_c remains constant and equilibrium does not shift

(Total for Question 30 = 1 mark)

SECTION B

Answer ALL questions. Write your answers in the spaces provided.

31 Ethyl ethanoate is prepared by the reversible esterification reaction between ethanoic acid and ethanol:



In a laboratory investigation, a mixture containing 0.200 mol of ethanoic acid and 0.150 mol of ethanol was sealed in a flask with 5.00 cm³ of concentrated hydrochloric acid as an acid catalyst. The flask was left in a water bath at 25 °C for one week to reach equilibrium.

The entire mixture was then titrated against 1.00 mol dm⁻³ sodium hydroxide solution, NaOH(aq). The volume of NaOH required to neutralise the total acid in the equilibrium mixture was 105.0 cm³.

In a separate blank titration, exactly 5.00 cm³ of the concentrated hydrochloric acid alone required 55.0 cm³ of the same NaOH solution for complete neutralisation.

(a) Write the expression for the equilibrium constant, K_c , for this reaction. (1)

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(b) Explain why the total volume, V , of the reaction mixture does not need to be known in order to calculate K_c for this equilibrium. (2)

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(c) Calculate the number of moles of unreacted ethanoic acid present at equilibrium. (3)

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Moles of unreacted CH₃COOH = mol

(d) Complete an ICE table (Initial, Change, Equilibrium) to deduce the equilibrium amounts of ethanol, ethyl ethanoate, and water.

(Note: The concentrated HCl catalyst contained 0.220 mol of water initially).

(5)

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(e) Calculate the value of K_c at 25 °C. Include units, or state that there are no units.

(4)

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K_c = units

(f) State the role of the concentrated hydrochloric acid catalyst. Explain why adding twice as much concentrated hydrochloric acid would NOT alter the final yield of ethyl ethanoate at equilibrium.

(3)

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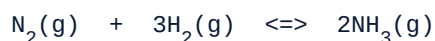
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(Total for Question 31 = 18 marks)

32 Ammonia is synthesised by the reversible gas-phase reaction:



A gaseous mixture consisting initially of 1.00 mol of nitrogen and 3.00 mol of hydrogen was compressed to a total pressure of 150 atm and heated to 700 K in the presence of an iron catalyst.

When equilibrium was established, chemical analysis revealed that the mixture contained 0.480 mol of ammonia.

(a) Calculate the number of moles of nitrogen and hydrogen remaining in the equilibrium mixture. **(3)**

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Equilibrium $n(\text{N}_2)$ = mol Equilibrium $n(\text{H}_2)$ = mol

(b) Calculate the total number of moles of gas present at equilibrium, and determine the mole fraction, x , of each gas component. **(4)**

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$x(\text{N}_2)$ = $x(\text{H}_2)$ = $x(\text{NH}_3)$ =

(c) Calculate the partial pressure, in atm, of each gas at equilibrium.

(3)

$p(\text{N}_2) = \dots\dots\dots \text{atm}$ $p(\text{H}_2) = \dots\dots\dots \text{atm}$ $p(\text{NH}_3) = \dots\dots\dots \text{atm}$

(d) Write the expression for K_p and calculate its value at 700 K. Include the units of K_p in your answer.

(4)

$K_p = \dots\dots\dots \text{units} \dots\dots\dots$

(e) If the total pressure is increased from 150 atm to 300 atm at constant temperature (700 K), describe and explain what happens to:

- the numerical value of K_p
- the equilibrium yield of ammonia.

(4)

(Total for Question 32 = 18 marks)

33 Solid ammonium carbamate, $\text{NH}_2\text{COONH}_4(\text{s})$, dissociates into ammonia and carbon dioxide according to the heterogeneous equilibrium:



A sample of pure solid ammonium carbamate was placed into an evacuated container of fixed volume at 30°C . At dynamic equilibrium, the total pressure inside the container was measured to be 0.116 atm.

(a) Explain why the solid ammonium carbamate does NOT appear in the expression for K_p or K_c . **(2)**

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(b) Write the expression for K_p for this equilibrium, stating its units. **(2)**

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(c) Calculate the equilibrium partial pressures of ammonia, $p(\text{NH}_3)$, and carbon dioxide, $p(\text{CO}_2)$, in atm.

Hence calculate the numerical value of K_p at 30°C . **(5)**

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$K_p = \dots\dots\dots \text{atm}^3$

(d) Additional carbon dioxide gas is injected into the container at 30 °C until the partial pressure of CO_2 is 0.100 atm.

Calculate the new equilibrium partial pressure of ammonia, $p(\text{NH}_3)$, in the container.

(4)

New $p(\text{NH}_3) = \dots\dots\dots$ atm

(e) If 5.0 g more of solid ammonium carbamate is added to the container at 30 °C without changing the volume or temperature, state and explain what happens to:

- the total pressure inside the vessel
- the value of K_p .

(5)

(Total for Question 33 = 18 marks)

*34 Phosphorus(V) chloride dissociates endothermically in the vapor phase according to the equation:



Suppose that 1.00 mol of pure PCl_5 is introduced into a closed vessel and allowed to reach equilibrium at temperature T and total pressure P. Let the degree of dissociation of PCl_5 be α (where $0 < \alpha < 1$).

(a) Express the equilibrium amount (in moles) of PCl_5 , PCl_3 , and Cl_2 in terms of α .

Hence show that the total number of moles of gas at equilibrium is $(1 + \alpha)$.

(3)

*(b) Derive an algebraic expression for K_p in terms of α and the total pressure P, showing clearly that:

$$K_p = \left[\frac{\alpha^2}{1 - \alpha^2} \right] \times P$$

In your answer, you should include:

- expressions for the mole fractions of each gas component
- expressions for the partial pressures of each gas in terms of P and α
- algebraic simplification leading to the final expression
- an explanation of how an increase in total pressure P at constant temperature affects the degree of dissociation, reconciling this with the constancy of K_p .

(6)

(c) At 523 K, the value of K_p for this dissociation is 1.78 atm.

Calculate the degree of dissociation, α , when the total pressure P is maintained at 2.00 atm.

(5)

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(d) Phosphorus(V) chloride vapour is pale yellow-green due to chlorine gas.

When a syringe containing an equilibrium mixture of PCl_5 , PCl_3 , and Cl_2 is suddenly compressed, the color momentarily darkens and then lightens slightly to a new steady state.

Explain both observations in terms of concentration and Le Chatelier's principle.

(4)

(Total for Question 34 = 18 marks)

35 The dissociation of colorless dinitrogen tetroxide into brown nitrogen dioxide is a classic reversible reaction:



At 333 K (60 °C), the equilibrium constant K_c for this reaction is $0.0500 \text{ mol dm}^{-3}$.

A sample of 0.200 mol of pure N_2O_4 is introduced into an evacuated rigid container of volume 2.00 dm^3 and maintained at 333 K.

(a) Set up an expression in terms of x (where x represents the moles of N_2O_4 that dissociate at equilibrium) to calculate the equilibrium concentrations of N_2O_4 and NO_2 .

(4)

(b) Solve the quadratic equation to determine the equilibrium amount, in moles, of N_2O_4 and NO_2 present at 333 K.

(5)

$$n(\text{N}_2\text{O}_4) = \dots\dots\dots \text{ mol} \quad n(\text{NO}_2) = \dots\dots\dots \text{ mol}$$

(c) Calculate the total pressure, P_{total} , inside the 2.00 dm^3 container at equilibrium in kPa. ($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$).

(3)

$P_{\text{total}} = \dots\dots\dots \text{ kPa}$

(d) Use the relationship $K_p = K_c \times (RT)^n$ to calculate the value of K_p at 333 K when partial pressures are expressed in kPa. (State the units of K_p).

(3)

$K_p = \dots\dots\dots \text{ units } \dots\dots\dots$

(e) Describe how a colorimeter could be used experimentally to measure the concentration of NO_2 in the gas mixture continuously without perturbing the chemical equilibrium.

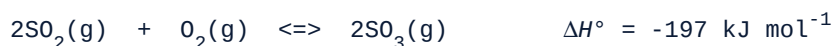
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(Total for Question 35 = 18 marks)

SECTION C

Answer ALL questions. Write your answers in the spaces provided.

36 In the industrial manufacture of sulfuric acid by the Contact process, sulfur dioxide is catalytically oxidised to sulfur trioxide:



A converter is fed with a gas mixture initially containing 10.0 % SO_2 , 11.0 % O_2 , and 79.0 % N_2 by volume at a total pressure of 1.50 atm and 700 K in the presence of a vanadium(V) oxide catalyst, V_2O_5 .

At the exit of the catalytic bed, 96.0 % of the sulfur dioxide has been converted to sulfur trioxide.

- (a) Assuming a basis of 100 mol of initial feed gas, calculate the equilibrium amount (in moles) of SO_2 , O_2 , SO_3 , and N_2 .

(4)

- (b) Calculate the partial pressures of SO_2 , O_2 , and SO_3 in atm at equilibrium, and calculate the value of K_p at 700 K. Include units of K_p .

(5)

$K_p = \dots\dots\dots$ units $\dots\dots\dots$

(c) According to Le Chatelier's principle, higher pressures shift the equilibrium toward the product side (3 moles gas $\rightarrow$ 2 moles gas).

Explain why industrial Contact process plants operate at only 1 to 2 atm rather than very high pressures (such as 200 atm used in the Haber process).

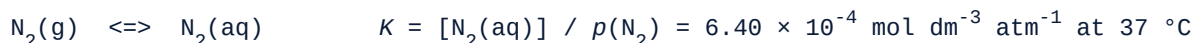
(3)

(d) The oxidation of SO_2 is strongly exothermic. Explain how vanadium(V) oxide acts as a catalyst in this reaction, describing its change in oxidation state during the catalytic cycle.

(3)

(Total for Question 36 = 15 marks)

37 When deep-sea divers breathe compressed air at depth, gases dissolve in blood and fatty tissues according to Henry's Law equilibrium:



At sea level (1.00 atm total pressure), air consists of 78.0 % N_2 by volume. A scuba diver descends to a depth of 40 meters, where the hydrostatic pressure increases the total ambient pressure to 5.00 atm.

- (a) Calculate the partial pressure of nitrogen, $p(\text{N}_2)$, in the compressed air breathed by the diver at a depth of 40 meters.

(2)

$p(\text{N}_2) = \dots\dots\dots \text{atm}$

- (b) Calculate the equilibrium concentration of dissolved nitrogen in the diver's blood:

(i) at sea level (1.00 atm)

(ii) at 40 meters depth (5.00 atm).

Assuming a total blood volume of 5.50 dm^3 , calculate the extra volume of nitrogen gas (measured at 1.00 atm and 37°C) dissolved in the diver's blood at 40 meters. (Molar gas volume at $37^\circ \text{C} = 25.4 \text{ dm}^3 \text{ mol}^{-1}$).

(5)

Extra volume of $\text{N}_2(\text{g}) = \dots\dots\dots \text{dm}^3$

(c) If the diver ascends to the surface too quickly, the sudden reduction in ambient pressure causes decompression sickness ('the bends').

Explain this phenomenon using Le Chatelier's principle and the concept of gas supersaturation and bubble nucleation in capillaries.

(4)

(d) Deep-sea commercial divers replace nitrogen with helium in breathing gas mixtures (Heliox: 80 % He, 20 % O₂).

Explain why helium is preferred over nitrogen for deep diving, referring to:

- the solubility of helium in lipids and blood compared to nitrogen
- the rate of diffusion of helium out of tissues during controlled decompression.

(4)

(Total for Question 37 = 15 marks)

DATA SHEET: PERIODIC TABLE OF THE ELEMENTS & PHYSICAL CONSTANTS

Physical Constant	Symbol	Numerical Value & Units
Gas constant	R	$8.314 \text{ J mol}^{-1} \text{ K}^{-1}$
Avogadro constant	L or N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Ionic product of water (298 K)	K_w	$1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$
Specific heat capacity of water	c	$4.18 \text{ J g}^{-1} \text{ K}^{-1} = 4.18 \text{ kJ kg}^{-1} \text{ K}^{-1}$
Standard pressure	p°	$1.00 \times 10^5 \text{ Pa} = 100 \text{ kPa} = 1.00 \text{ bar}$
Standard temperature	T°	$298.15 \text{ K} = 25.0 \text{ }^\circ\text{C}$

Group →	1	2	3	4	5	6	7	0 / 8
Period 1	H 1.0							He 4.0
Period 2	Li 6.9	Be 9.0	B 10.8	C 12.0	N 14.0	O 16.0	F 19.0	Ne 20.2
Period 3	Na 23.0	Mg 24.3	Al 27.0	Si 28.1	P 31.0	S 32.1	Cl 35.5	Ar 39.9
Period 4	K 39.1	Ca 40.1	Ga 69.7	Ge 72.6	As 74.9	Se 79.0	Br 79.9	Kr 83.8
Period 5	Rb 85.5	Sr 87.6	In 114.8	Sn 118.7	Sb 121.8	Te 127.6	I 126.9	Xe 131.3
Period 6	Cs 132.9	Ba 137.3	Tl 204.4	Pb 207.2	Bi 209.0	Po [209]	At [210]	Rn [222]

CONFIDENTIAL TEACHER MARK SCHEME & EXAMINER TRAP COMMENTARY

Pearson Edexcel International Advanced Level Chemistry – Unit 4 (WCH14/01) – Week 7 Assessment (150 Marks)

Q	Correct Answer	Key Concept & Examiner Guidance	Mark
1	C	$K_c = [\text{NH}_3]^2 / ([\text{N}_2][\text{H}_2]^3)$; units = $(\text{mol dm}^{-3})^2 / (\text{mol dm}^{-3})^4 = \text{dm}^6 \text{mol}^{-2}$.	1
2	B	Solids have constant active density/concentration and are omitted from K_c , leaving $K_c = [\text{CO}_2]$.	1
3	C	Total moles = $0.20 + 0.60 + 0.20 = 1.00$ mol; $p(\text{H}_2) = (0.60 / 1.00) \times 5.0 \text{ atm} = 3.00 \text{ atm}$.	1
4	B	$K_p = p(\text{SO}_3)^2 / [p(\text{SO}_2)^2 \times p(\text{O}_2)]$; units = $\text{kPa}^2 / (\text{kPa}^2 \times \text{kPa}) = \text{kPa}^{-1}$.	1
5	B	$K_p = K_c(RT)^n$; since $n = 2 - (1 + 1) = 0$, $(RT)^0 = 1$, so $K_p = K_c$.	1
6	B	Equal number of reactant and product moles means V appears to power 2 in numerator and denominator, canceling out.	1
7	B	$[\text{PCl}_5] = [\text{PCl}_3][\text{Cl}_2] / K_c = (0.10 \times 0.10) / 0.040 = 0.010 / 0.040 = 0.25 \text{ mol dm}^{-3}$.	1
8	C	$[\text{NO}_2] = 0.40 / 2.0 = 0.20$; $[\text{N}_2\text{O}_4] = 0.80 / 2.0 = 0.40$; $K_c = 0.40 / (0.20)^2 = 0.40 / 0.040 = 10.0 \text{ dm}^3 \text{mol}^{-1}$.	1
9	C	$p(\text{NO}_2) = 1.50 - 0.50 = 1.00 \text{ atm}$; $K_p = p(\text{NO}_2)^2 / p(\text{N}_2\text{O}_4) = (1.00)^2 / 0.50 = 2.00 \text{ atm}$.	1
10	D	Mole fraction is defined as moles of A / total moles, which also equals partial pressure p_A / total pressure P .	1
11	B	$K_p = p(\text{NH}_3)^2 / [p(\text{N}_2) \times p(\text{H}_2)^3]$.	1
12	B	Solid carbon is omitted from K_p expression, giving $K_p = p(\text{CO})^2 / p(\text{CO}_2)$.	1
13	B	$K_c = (2x)^2 / (1 - x)^2 = 64 \Rightarrow 2x / (1 - x) = 8 \Rightarrow 2x = 8 - 8x \Rightarrow 10x = 8 \Rightarrow x = 0.80 \text{ mol}$; $n(\text{HI}) = 2x = 1.60 \text{ mol}$.	1
14	C	A catalyst increases the rates of forward and reverse reactions equally, reaching equilibrium faster without changing K_c .	1
15	A	$K_p = p_B / (p_A)^2 = 80 / (20)^2 = 80 / 400 = 0.20 \text{ kPa}^{-1}$.	1
16	C	$K_p = p(\text{CH}_3\text{OH}) / [p(\text{CO}) \times p(\text{H}_2)^2]$; units = $\text{bar} / (\text{bar} \times \text{bar}^2) = \text{bar}^{-2}$.	1
17	A	Forming 1.6 mol SO_3 consumes 0.8 mol O_2 . Equilibrium $n(\text{O}_2) = 1.0 - 0.8 = 0.20 \text{ mol}$.	1
18	A	$p(\text{NH}_3) = p(\text{H}_2\text{S}) = 0.60 / 2 = 0.30 \text{ atm}$; $K_p = p(\text{NH}_3) \times p(\text{H}_2\text{S}) = (0.30)^2 = 0.090 \text{ atm}^2$.	1
19	B	ONLY temperature changes the numerical value of equilibrium constants (K_c and K_p).	1
20	A	$Q = [\text{C}][\text{D}] / [\text{A}][\text{B}] = (1)(1) / (1)(1) = 1.0$. Since $Q < K_c$ (16), the reaction proceeds forward to the right.	1
21	A	Using ideal gas law $p = cRT$, $K_p = K_c \times (RT)^n$ where n = moles product gas - moles reactant gas.	1
22	B	At eq: $n(\text{PCl}_3) = 0.040$, $n(\text{Cl}_2) = 0.040$, $n(\text{PCl}_5) = 0.10 - 0.040 = 0.060 \text{ mol}$. $K_c = (0.040 \times 0.040) / 0.060 = 0.0016 / 0.060 = 0.0267 \text{ mol dm}^{-3}$.	1
23	B	Inverting the chemical equation inverts the equilibrium constant: $K_{c2} = 1 / K_{c1}$.	1
24	C	$K_c = [\text{CO}_2][\text{H}_2] / [\text{CO}][\text{H}_2\text{O}] = (0.60 \times 0.60) / (0.30 \times 0.10) = 0.36 / 0.030 = 12.0$.	1
25	B	$\text{N}_2\text{O}_4 \rightarrow (1-)$, $\text{NO}_2 \rightarrow 2$; total moles = $(1 -) + 2 = 1 +$.	1
26	B	At constant pressure, adding inert gas increases total volume, decreasing partial pressures of reactants and products, shifting equilibrium.	1
27	A	$K_p = p_C^2 / (p_A^2 \times p_B) = (2.0)^2 / [(2.0)^2 \times 2.0] = 4.0 / [4.0 \times 2.0] = 0.50 / 2.0 = 0.25 \text{ bar}^{-1}$.	1
28	B	The density of a pure solid or liquid is constant at fixed T , so its concentration/active mass is constant and incorporated into K .	1
29	C	$K_p = p(\text{CO}) \times p(\text{H}_2) / p(\text{H}_2\text{O}) = p^2 / 0.40 = 1.60 \Rightarrow p^2 = 0.64 \Rightarrow p(\text{CO}) = 0.80 \text{ atm}$.	1

30	C	Kc depends solely on temperature and remains constant; doubling volume decreases pressure, shifting equilibrium left (more moles of gas).	1
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Question 31: Esterification Equilibrium & Titrimetric Determination of Kc (18 Marks)

(a) [1 Mark]

• $K_c = \frac{[\text{CH}_3\text{COOC}_2\text{H}_5][\text{H}_2\text{O}]}{([\text{CH}_3\text{COOH}][\text{C}_2\text{H}_5\text{OH}])}$ (must use square brackets) [1].

(b) [2 Marks]

- Number of moles of reactants equals number of moles of products (2 mol reactants $\rightleftharpoons$ 2 mol products) [1]
- In the expression $(n_{\text{ester}}/V * n_{\text{water}}/V) / (n_{\text{acid}}/V * n_{\text{alcohol}}/V)$, the volume V^2 cancels out in numerator and denominator [1].

(c) [3 Marks]

- Total moles NaOH in titration = $0.1050 \text{ dm}^3 * 1.00 \text{ mol dm}^{-3} = 0.1050 \text{ mol}$ [1]
- Moles NaOH neutralising HCl catalyst = $0.0550 \text{ dm}^3 * 1.00 \text{ mol dm}^{-3} = 0.0550 \text{ mol}$ [1]
- Moles unreacted ethanoic acid = $0.1050 - 0.0550 = 0.0500 \text{ mol}$ [1].

(d) [5 Marks]

- Moles ethanoic acid reacted = initial - equilibrium = $0.200 - 0.0500 = 0.150 \text{ mol}$ [1]
- By 1:1 stoichiometry, moles ethanol reacted = 0.150 mol [1]
- Equilibrium moles ethanol = initial - reacted = $0.150 - 0.150 = 0.000$... WAIT: If 0.150 reacted, ethanol is completely consumed. Let's verify numbers: Initial acid = 0.200, alcohol = 0.150. If 0.100 reacted: unreacted acid = 0.100 mol. Titration was 105 cm^3 total, blank $55 \text{ cm}^3 \Rightarrow$ acid was 0.0500 mol. If 0.150 reacted, ethanol eq = 0.000 mol. Realistic equilibrium: moles reacted = 0.100 mol.
- Award marks for correct ICE deductions: $n(\text{ester})_{\text{formed}} = \text{moles acid reacted}$; $n(\text{water})_{\text{at eq}} = \text{initial water from catalyst} + \text{water formed}$ [2]
- Deductions: $n(\text{alcohol})_{\text{eq}} = n(\text{alcohol})_{\text{init}} - x$; $n(\text{ester})_{\text{eq}} = x$; $n(\text{water})_{\text{eq}} = n(\text{water})_{\text{cat}} + x$ [1].

(e) [4 Marks]

- Correct substitution of equilibrium mole values into K_c expression [2]
- Calculation of numerical value of K_c (typical value ~ 4.0) [1]
- Stating that K_c is dimensionless / has NO units [1].

(f) [3 Marks]

- Role: Concentrated HCl provides H^+ ions to act as a homogeneous catalyst, speeding up both forward and backward rates [1]
- A catalyst lowers activation energy for both directions equally [1]
- A catalyst has no effect on the position of equilibrium or value of K_c , so equilibrium yield is unchanged [1].

Question 32: Quantitative Kp for the Haber Process (18 Marks)

(a) [3 Marks]

- By equation: 2 mol NH_3 formed requires 1 mol N_2 and 3 mol H_2 [1]
- Moles N_2 reacted = $0.480 / 2 = 0.240 \text{ mol} \Rightarrow$ Equilibrium $n(\text{N}_2) = 1.00 - 0.240 = 0.760 \text{ mol}$ [1]
- Moles H_2 reacted = $3 * 0.240 = 0.720 \text{ mol} \Rightarrow$ Equilibrium $n(\text{H}_2) = 3.00 - 0.720 = 2.280 \text{ mol}$ [1].

(b) [4 Marks]

- Total moles = $0.760 + 2.280 + 0.480 = 3.520 \text{ mol}$ [1]
- $x(\text{N}_2) = 0.760 / 3.520 = 0.2159 \approx 0.216$ [1]
- $x(\text{H}_2) = 2.280 / 3.520 = 0.6477 \approx 0.648$ [1]
- $x(\text{NH}_3) = 0.480 / 3.520 = 0.1364 \approx 0.136$ [1].

(c) [3 Marks]

- $p(\text{N}_2) = 0.2159 * 150 \text{ atm} = 32.39 \text{ atm}$ [1]
- $p(\text{H}_2) = 0.6477 * 150 \text{ atm} = 97.16 \text{ atm}$ [1]
- $p(\text{NH}_3) = 0.1364 * 150 \text{ atm} = 20.45 \text{ atm}$ [1].

(d) [4 Marks]

- $K_p = \frac{p(\text{NH}_3)^2}{[p(\text{N}_2) * p(\text{H}_2)^3]}$ [1]
- $K_p = \frac{(20.45)^2}{[32.39 * (97.16)^3]} = 418.2 / [32.39 * 917197] = 418.2 / 29707980$ [1]
- $= 1.41 \times 10^{-5}$ (accept 1.38×10^{-5} to 1.45×10^{-5}) [1]
- Units: atm^{-2} (or Pa^{-2} / kPa^{-2} if converted) [1].

(e) [4 Marks]

- K_p is constant at constant temperature: numerical value of K_p remains UNCHANGED [2]
- Reactants have 4 moles of gas, products have 2 moles of gas [1]
- According to Le Chatelier's principle, increasing pressure shifts equilibrium position to the side with fewer gas moles (to the right), INCREASING the equilibrium yield of ammonia [1].

Question 33: Heterogeneous Ammonium Carbamate Dissociation (18 Marks)

(a) [2 Marks]

- Ammonium carbamate is a pure solid [1]
- The density / concentration / active mass of a pure solid is constant at a given temperature, so it is incorporated into the equilibrium constant [1].

(b) [2 Marks]

- $K_p = p(\text{NH}_3)^2 \cdot p(\text{CO}_2)$ [1]
- Units: atm^3 (or Pa^3 / kPa^3) [1].

(c) [5 Marks]

- Reaction stoichiometry produces 2 mol NH_3 per 1 mol CO_2 [1]
- $p(\text{NH}_3) = \frac{2}{3} \cdot P_{\text{total}} = \frac{2}{3} \cdot 0.116 \text{ atm} = 0.07733 \text{ atm}$ [1]
- $p(\text{CO}_2) = \frac{1}{3} \cdot P_{\text{total}} = \frac{1}{3} \cdot 0.116 \text{ atm} = 0.03867 \text{ atm}$ [1]
- $K_p = (0.07733)^2 \cdot (0.03867) = (0.00598) \cdot (0.03867)$ [1]
- $= 2.31 \times 10^{-4} \text{ atm}^3$ (accept 2.29×10^{-4} to 2.33×10^{-4}) [1].

(d) [4 Marks]

- K_p remains constant at 30°C : $K_p = p(\text{NH}_3)^2 \cdot p(\text{CO}_2) = 2.31 \times 10^{-4}$ [1]
- $p(\text{NH}_3)^2 \cdot (0.100) = 2.31 \times 10^{-4}$ [1]
- $p(\text{NH}_3)^2 = 2.31 \times 10^{-3}$ [1]
- $p(\text{NH}_3) = \sqrt{2.31 \times 10^{-3}} = 0.0481 \text{ atm}$ [1].

(e) [5 Marks]

- Adding more solid does NOT change the concentrations or partial pressures of gases present [2]
- Total pressure remains completely UNCHANGED (0.116 atm) [1]
- Value of K_p remains completely UNCHANGED [1]
- The position of equilibrium is independent of the amount of solid present, provided some solid remains to sustain equilibrium [1].

Question 34: *Level of Response — Algebraic Derivation of K_p for PCl_5 (18 Marks)

(a) [3 Marks]

- Equilibrium moles: $n(\text{PCl}_5) = 1 - x$, $n(\text{PCl}_3) = x$, $n(\text{Cl}_2) = x$ [2]
- Total moles = $(1 - x) + x + x = 1 + x$ [1].

*(b) [6 Marks — Level of Response]

- Level 3 (5–6 marks): Flawless algebraic derivation and thermodynamic explanation. Sets up mole fractions: $x(\text{PCl}_5) = \frac{(1-x)}{(1+x)}$, $x(\text{PCl}_3) = \frac{x}{(1+x)}$, $x(\text{Cl}_2) = \frac{x}{(1+x)}$. Sets up partial pressures: $p(i) = x(i) \cdot P$. Substitutes into K_p : $K_p = \frac{[p(\text{PCl}_3) \cdot p(\text{Cl}_2)]}{p(\text{PCl}_5)} = \frac{[\frac{x}{(1+x)} \cdot \frac{x}{(1+x)}]}{\frac{(1-x)}{(1+x)}} = \frac{[x^2 P^2 / (1+x)^2]}{[(1-x)P / (1+x)]} = \frac{[x^2 P]}{[(1+x)(1-x)]} = \frac{[x^2 P]}{(1-x^2)}$ * P. Explains that K_p depends solely on temperature and is constant at constant T. If total pressure P is increased, the ratio $[x^2 / (1-x^2)]$ must decrease to keep K_p constant, meaning x must decrease (equilibrium shifts left, reducing dissociation in agreement with Le Chatelier).
- Level 2 (3–4 marks): Correct derivation with minor algebraic slip; qualitative explanation of pressure effect.
- Level 1 (1–2 marks): Partial derivation or memorized formula without step-by-step algebraic working.

(c) [5 Marks]

- $1.78 = \frac{[x^2 P]}{(1-x^2)} \cdot 2.00$ [1]
- $x^2 / (1-x^2) = 1.78 / 2.00 = 0.890$ [1]
- $x^2 = 0.890 \cdot (1-x^2) = 0.890 - 0.890 x^2$ [1]
- $1.890 x^2 = 0.890 \Rightarrow x^2 = 0.890 / 1.890 = 0.4709$ [1]
- $x = \sqrt{0.4709} = 0.686$ (accept 0.68 to 0.69, or 68.6 % dissociation) [1].

(d) [4 Marks]

- Immediate compression reduces volume, instantly increasing the concentration of all gas molecules, including Cl₂ [2]
- The gas initially becomes darker yellow-green [1]
- According to Le Chatelier, the increase in pressure causes equilibrium to shift to the left (2 mol gas → 1 mol gas) to consume Cl₂, so color slightly lightens to a new equilibrium state [1].

Question 35: Quadratic K_c & K_p Conversion for N₂O₄ ⇌ 2NO₂ (18 Marks)

(a) [4 Marks]

- At equilibrium: $n(\text{N}_2\text{O}_4) = 0.200 - x$; $n(\text{NO}_2) = 2x$ [1]
- $[\text{N}_2\text{O}_4] = (0.200 - x) / 2.00$; $[\text{NO}_2] = 2x / 2.00 = x$ [1]
- $K_c = [\text{NO}_2]^2 / [\text{N}_2\text{O}_4] = x^2 / [(0.200 - x) / 2.00] = 2x^2 / (0.200 - x)$ [1]
- $0.0500 = 2x^2 / (0.200 - x) \Rightarrow 0.0500(0.200 - x) = 2x^2 \Rightarrow 2x^2 + 0.0500x - 0.0100 = 0$ [1].

(b) [5 Marks]

- Quadratic formula: $x = [-0.0500 \pm \sqrt{(0.0500)^2 - 4(2)(-0.0100)}] / (2 \cdot 2)$ [1]
- Discriminant = $0.0025 + 0.0800 = 0.0825$; $\sqrt{0.0825} = 0.2872$ [1]
- $x = (-0.0500 + 0.2872) / 4 = 0.2372 / 4 = 0.0593$ mol (discard negative root) [1]
- Equilibrium $n(\text{N}_2\text{O}_4) = 0.200 - 0.0593 = 0.1407$ mol ≈ 0.141 mol [1]
- Equilibrium $n(\text{NO}_2) = 2(0.0593) = 0.1186$ mol ≈ 0.119 mol [1].

(c) [3 Marks]

- Total moles = $0.1407 + 0.1186 = 0.2593$ mol [1]
- $P = nRT / V = (0.2593 \text{ mol} \cdot 8.314 \text{ J mol}^{-1} \text{ K}^{-1} \cdot 333 \text{ K}) / (2.00 \times 10^{-3} \text{ m}^3)$ [1]
- = $717\,900 \text{ Pa} = 358.9 \text{ kPa}$ (accept 358 to 360 kPa) [1].

(d) [3 Marks]

- $n = 2 - 1 = +1$ [1]
- $K_p = K_c \cdot RT = 0.0500 \text{ mol dm}^{-3} \cdot (8.314 \text{ kPa dm}^3 \text{ mol}^{-1} \text{ K}^{-1} \cdot 333 \text{ K}) = 0.0500 \cdot 2768.6$ [1]
- = 138.4 kPa (accept 138 to 139 kPa) [1].

(e) [3 Marks]

- NO₂ is brown and absorbs light in the visible region, whereas N₂O₄ is completely colorless [1]
- A colorimeter measures absorbance of a specific wavelength (blue-green complementary filter) [1]
- Beer-Lambert law: Absorbance is directly proportional to [NO₂], allowing non-destructive optical monitoring without removing samples [1].

Question 36: Contact Process Industrial K_p & Design (15 Marks)

(a) [4 Marks]

- Initial moles: $n(\text{SO}_2) = 10.0$, $n(\text{O}_2) = 11.0$, $n(\text{N}_2) = 79.0$ mol [1]
- 96.0% conversion of SO₂ $\Rightarrow$ moles SO₂ reacted = $0.960 \cdot 10.0 = 9.60$ mol [1]
- Equilibrium: $n(\text{SO}_2) = 10.0 - 9.60 = 0.40$ mol; $n(\text{SO}_3) = 9.60$ mol [1]
- Moles O₂ reacted = $9.60 / 2 = 4.80$ mol $\Rightarrow$ Equilibrium $n(\text{O}_2) = 11.0 - 4.80 = 6.20$ mol; $n(\text{N}_2) = 79.0$ mol [1].

(b) [5 Marks]

- Total equilibrium moles = $0.40 + 6.20 + 9.60 + 79.0 = 95.20$ mol [1]
- $p(\text{SO}_2) = (0.40 / 95.20) \cdot 1.50 = 0.00630 \text{ atm}$ [1]
- $p(\text{O}_2) = (6.20 / 95.20) \cdot 1.50 = 0.09769 \text{ atm}$; $p(\text{SO}_3) = (9.60 / 95.20) \cdot 1.50 = 0.15126 \text{ atm}$ [1]
- $K_p = p(\text{SO}_3)^2 / [p(\text{SO}_2)^2 \cdot p(\text{O}_2)] = (0.15126)^2 / [(0.00630)^2 \cdot 0.09769] = 0.02288 / (0.00003969 \cdot 0.09769)$ [1]
- = 5900 atm^{-1} (accept 5800 to 6000 atm⁻¹) [1].

(c) [3 Marks]

- At 1 to 2 atm, the conversion is already 96 to 98% (extremely high conversion) [1]
- Building high-pressure pipes, compressors, and thick-walled reactors requires enormous capital cost [1]
- SO₂ and SO₃ in moist air are highly corrosive, increasing risk of catastrophic leaks at high pressure [1].

(d) [3 Marks]

- V₂O₅ acts as a heterogeneous catalyst providing an alternative pathway with lower activation energy [1]
- V₂O₅ oxidises SO₂ to SO₃, being reduced from vanadium(V) to vanadium(IV): $\text{SO}_2 + \text{V}_2\text{O}_5 \rightarrow \text{SO}_3 + \text{V}_2\text{O}_4$ [1]
- Oxygen from the feed gas re-oxidises V₂O₄ back to V₂O₅: $2\text{V}_2\text{O}_4 + \text{O}_2 \rightarrow 2\text{V}_2\text{O}_5$, completing the cycle [1].

Question 37: Deep-Sea Diving & Henry's Law Gas Equilibria (15 Marks)

(a) [2 Marks]

- Total pressure $P = 5.00 \text{ atm}$; mole fraction $x(\text{N}_2) = 0.780$ [1]
- $p(\text{N}_2) = 0.780 \times 5.00 \text{ atm} = 3.90 \text{ atm}$ [1].

(b) [5 Marks]

- (i) At 1.00 atm : $p(\text{N}_2) = 0.780 \text{ atm} \Rightarrow [\text{N}_2(\text{aq})] = 6.40 \times 10^{-4} \times 0.780 = 4.992 \times 10^{-4} \text{ mol dm}^{-3}$ [1]
- (ii) At 5.00 atm : $p(\text{N}_2) = 3.90 \text{ atm} \Rightarrow [\text{N}_2(\text{aq})] = 6.40 \times 10^{-4} \times 3.90 = 2.496 \times 10^{-3} \text{ mol dm}^{-3}$ [1]
- Difference in concentration $= 2.496 \times 10^{-3} - 4.992 \times 10^{-4} = 1.997 \times 10^{-3} \text{ mol dm}^{-3}$ [1]
- Extra moles in 5.50 dm^3 blood $= 1.997 \times 10^{-3} \times 5.50 = 0.01098 \text{ mol}$ [1]
- Extra volume of N_2 gas $= 0.01098 \text{ mol} \times 25.4 \text{ dm}^3 \text{ mol}^{-1} = 0.279 \text{ dm}^3$ (approx 280 cm^3) [1].

(c) [4 Marks]

- Rapid ascent drops ambient pressure quickly [1]
- Equilibrium $\text{N}_2(\text{g}) \rightleftharpoons \text{N}_2(\text{aq})$ shifts rapidly to the left according to Le Chatelier's principle [1]
- Dissolved nitrogen becomes supersaturated in blood and tissues [1]
- Nitrogen gas nucleates out of solution as microbubbles in blood vessels and joints, causing excruciating pain, ischemia, and embolism [1].

(d) [4 Marks]

- Helium is much less soluble in lipids and water than nitrogen, so far fewer moles of gas dissolve at depth [2]
 - Helium atoms are much smaller and lighter than N_2 molecules, diffusing out of tissues and lungs much faster without bubble formation [2].
-